CIE AS PHYSICS (9702) • QUESTION PAPER

5.1 Energy Conservation

Syllabus Section: 5. Work, Energy & Power **Paper Type:** Structured Questions (Paper 2) **Total:** 11 Questions (123 Marks)

Question 1 Easy [11 marks]

Part (a) [2 marks]

A small child does work on a large suitcase by pushing it along a smooth floor, as shown in Fig. 1.1.



Fig. 1.1.

State the definition of work done and write the equation.

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.....

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Part (b) [2 marks]

The suitcase is moved back to its original position by an adult who pulls it at an angle of 70° to the floor, as shown in Fig. 1.2.

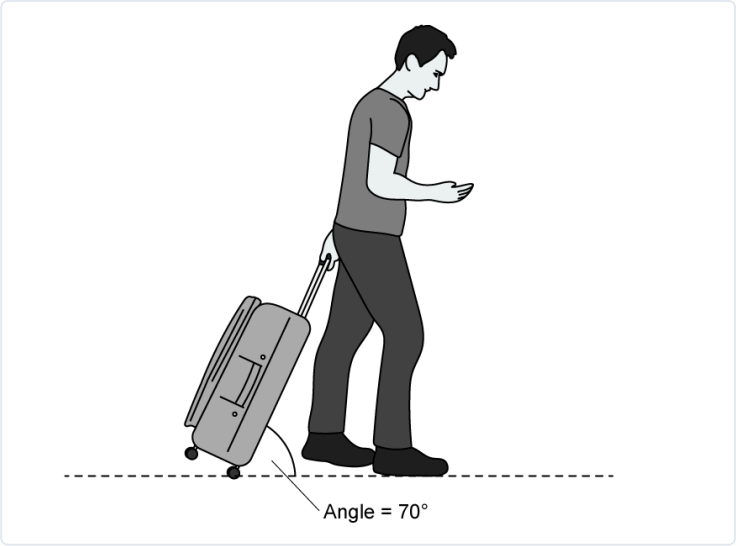


Fig. 1.2.

Write an expression for the work done on the suitcase by the adult in terms of force F, distance s and the angle 70°.

Part (c) **[4 marks]**

In moving the suitcase the child and adult both exerted equal forces of 20 N. They each move the suitcase a distance of 2.5 m.

Calculate the work done by

(i) the child

[2]

(ii) the adult.

[2]

Part (d) **[3 marks]**

State whether the child or the adult are more efficient at moving the suitcase, explaining your reasoning.

Question 2 Easy [6 marks]

Part (a) [2 marks]

A block of weight 4.25 N is placed on a frictionless slope inclined at 30° to the horizontal as shown in Fig. 1.1.

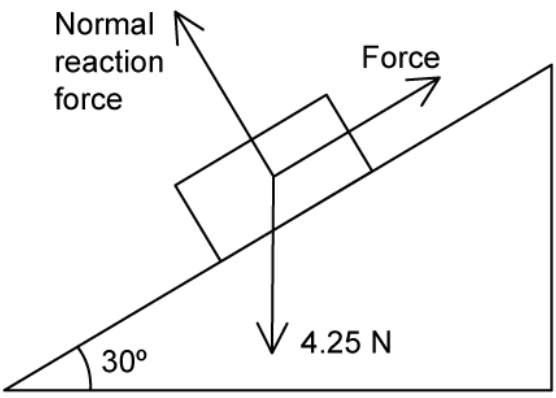


Fig. 1.1.

The block is moved up the slope at constant speed by applying a force parallel to the slope.

Calculate the component of the weight of the block which acts parallel to the slope.

Part (b) [2 marks]

Calculate the work done in moving the block a distance of 6.0 m up the slope.

Part (c) [2 marks]

The block in part (a) is pushed 6.0 m up the slope in 18 s.

Calculate the power needed to move the block in this amount of time.

Question 3 Easy [5 marks]

Part (a) [2 marks]

A beaker is filled with ice and then an electric heater placed into it, as shown in Fig. 1.1. The heater is turned on and heats up until it glows red.

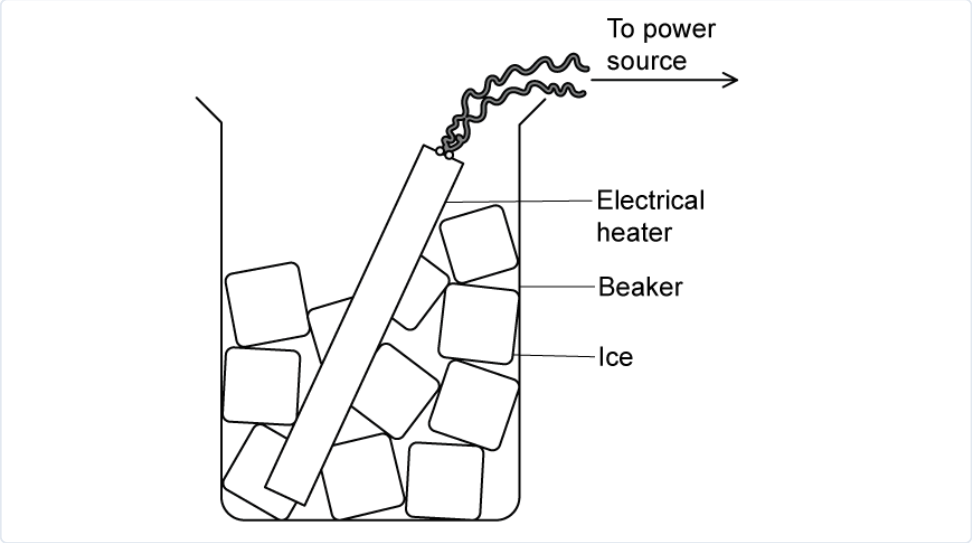


Fig. 1.1.

A student states that when the ice is being heated up the total amount of energy in the ice remains the same.

State the principle of conservation of energy.

Part (b) [3 marks]

Explain whether the student is correct, giving your reason.

Question 4 Medium [12 marks]

Part (a) [1 mark]

Define work done by a force.

Part (b) [2 marks]

A heavy bag is pulled across a rough surface at a constant speed.

Describe and explain how work is done in this situation.

Part (c) [3 marks]

A crane is used to lift the bag from the ground onto the loading area of a truck.

Fig. 1.1 shows the crane moving the bag from the ground to the truck.

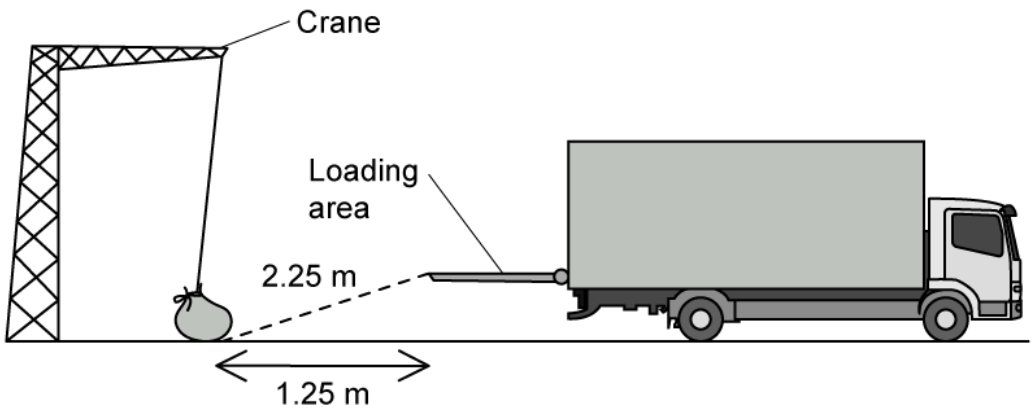


Fig. 1.1

The mass of the bag is 64 kg. The crane takes 0.1 minutes to lift the bag from the ground to the loading area of the truck.

Calculate the rate of work done by the crane when moving the bag.

Part (d) **[6 marks]**

The crane is powered using an electric motor. The efficiency of the crane is 74 %.

The crane is used to load approximately 400 bags with an average mass of 60 kg each day.

(i) Calculate the power used by the crane's motor to lift the bag in part (c).

[1]

(ii) The typical working efficiency of a crane is 80%.

Explain how the difference in efficiency may affect the number of bags the crane can lift in a day.

[3]

(iii) Suggest **one** change which could increase the efficiency of the motor and explain how it would achieve this.

[2]

Question 5 Medium [9 marks]

Part (a) [1 mark]

Engineers around the world are working on ways to provide enough energy to sustain human activity.

Off-shore wind farms use the kinetic energy of strong offshore winds to generate electricity.

Fig. 1.1 shows an offshore wind power installation.

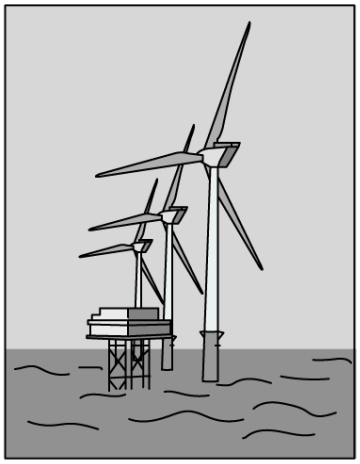


Fig. 1.1

State the definition of power and give the units used to measure power output.

Part (b) [3 marks]

The UK is a world leader in the installation of offshore wind power.

Located in the North Sea off the east coast of England, the Hornsea Phase 2 (Hornsea Two) offshore wind farm was the largest wind farm in the world when it opened in 2022.

It uses 165 turbines rated at 8 MW each, to generate 1.4 GW of energy, enough to supply more than 1.3 million homes. Each turbine has blades of length 81 m.

The power output of a single turbine can be calculated using

$$P = \frac{1}{2} \rho A v^3$$

where ρ is the density of the air, A is the area swept out by the blades and v is the wind speed.

Calculate the average wind speed at Hornsea Two.

The density of air is 1.225 kg m^{-3}.

Part (c) **[5 marks]**

The Hornsea wind turbines are found to have an efficiency of 40%.

(i) Calculate the energy required from the wind every second to achieve the desired power output.

[2]

(ii) Explain why it is impossible to transfer all the energy available from the wind.

[3]

Question 6 Medium [17 marks]

Part (a) [5 marks]

A hybrid electric bike is fitted with both pedals and a battery. The cyclist can use pedals only, pedals and battery, or battery only as they ride. Fig 1.1 shows a cyclist riding a hybrid electric bike up an inclined road.

At one hill the cyclist switches to battery only. A road sign at the base of the hill states that it has an incline of 5%, meaning that for every 100 m travelled horizontally, the height increases by 5 m.

The bike travels at a steady speed of 6.0 m s^{-1}.

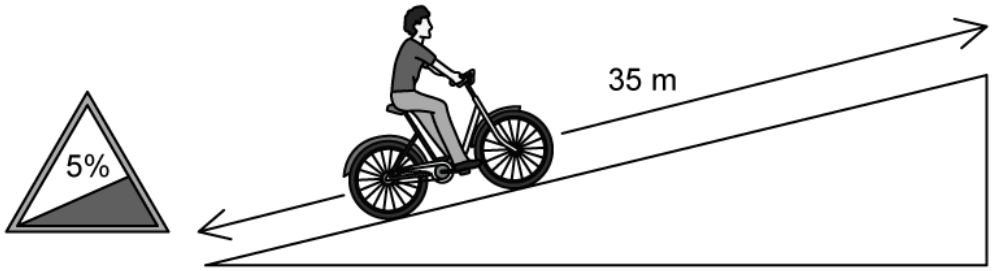


Fig. 1.1 (not to scale)

The mass of the rider and bike combined is 72.8 kg. The distance from the base to the top of the hill is 35 m.

Calculate the minimum power required from the battery. Assume that there is no energy loss due to friction.

Part (b) [5 marks]

Near the top of the hill, the cyclist stops pedalling and the bicycle freewheels for 1.6 s until coming to a stop.

The frictional forces which slow down the motion are constant.

(i) Show that the bicycle travels about 5 m before coming to a stop.

[2]

(ii) Calculate the magnitude of the frictional force which slows the bicycle down.

[3]

Part (c) **[7 marks]**

Once the cyclist reaches the top, they turn off the motor and head back down to the bottom. At the bottom of the hill, they reach a velocity of 8.8 m s^{-1}.

(i) Calculate the energy the cyclist has at the top of the hill.

[2]

(ii) Calculate the energy the cyclist has at the bottom of the hill.

[2]

(iii) Compare the two values you have calculated and suggest a reason for the difference.

[3]

Question 7 Medium [13 marks]

Part (a) [3 marks]

Fig. 1.1 shows a ride at a water park. Customers are seated in a circular glider which is subject to an accelerating force at point **A**.

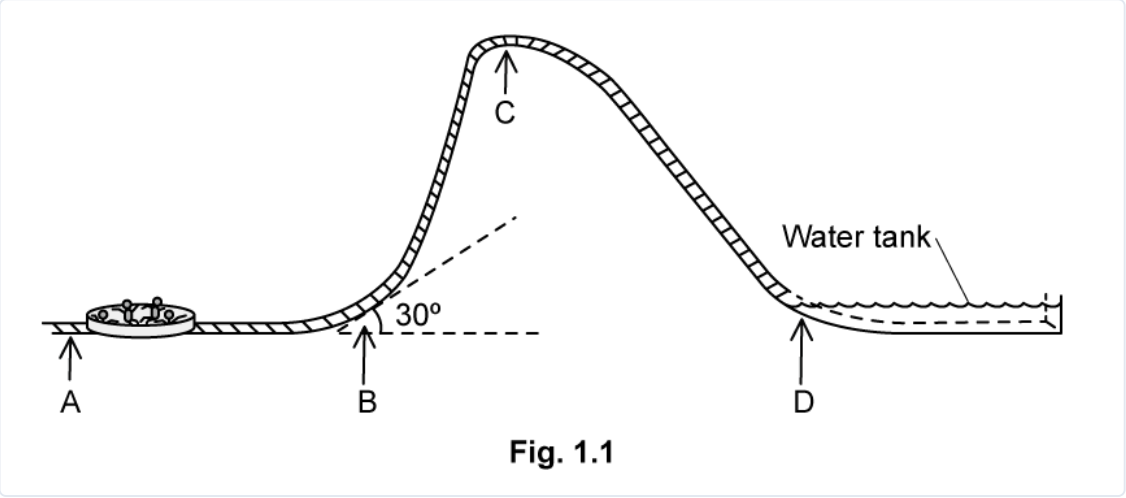


Fig. 1.1

The first part of the ride is horizontal. In this section, the glider is accelerated from rest at point **A** to 30 m s^{-1} at point **B**.

At point **B**, the track is angled at 30° to the horizontal for 2.5 m.

The total mass of the glider and passengers is 350 kg.

Calculate the work done by the glider against gravity when it travels through point **B**.

Part (b) [4 marks]

(i) Calculate the maximum height above **A** that the passengers and glider could reach.

[3]

(ii) Point **C** is 28 m above **A**.

Suggest why the ride is built to this height instead of the maximum height calculated in **(i)**.

[1]

Part (c) [3 marks]

At point **C**, the glider stops momentarily before sliding down the incline until it lands in the water pool at point **D** with a speed of 22 m s^{-1}.

Determine the efficiency of the energy transfer between points **C** and **D**.

Part (d) [3 marks]

Once the glider reaches point **D**, the water applies a decelerating force until it comes to a stop after 8.5 m.

Calculate the decelerating force exerted on the glider by the water.

State any assumptions you made in your calculation.

Question 8 Medium [16 marks]

Part (a) [7 marks]

Fig. 1.1 shows a simple arrangement which can be used to investigate how the kinetic energy of a trolley varies with its distance from the top of the ramp.

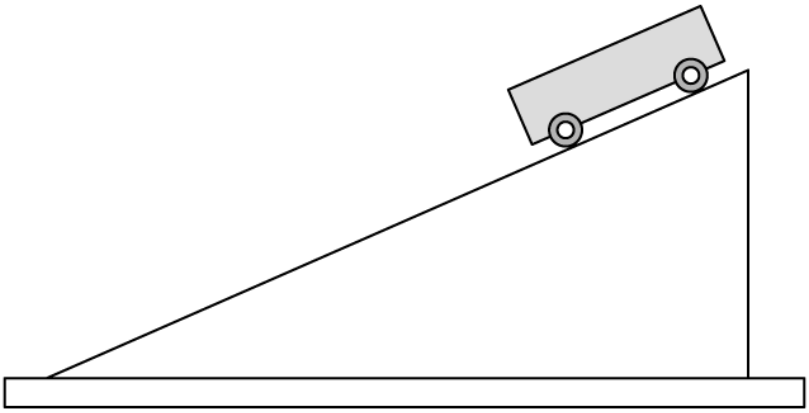


Fig. 1.1

In order to plan an investigation to determine the kinetic energy of the car at a particular point on the ramp:

(i) Describe briefly any additional equipment and the method to be used.

[3]

(ii) State the measurements to be taken and the measuring tool in each case.

[2]

(iii) Explain how the data will be analysed.

[2]

Part (b) [4 marks]

The graph in Fig 1.2 shows the variation of the gravitational potential energy and kinetic energy of the trolley with distance *d* from the top of the ramp.

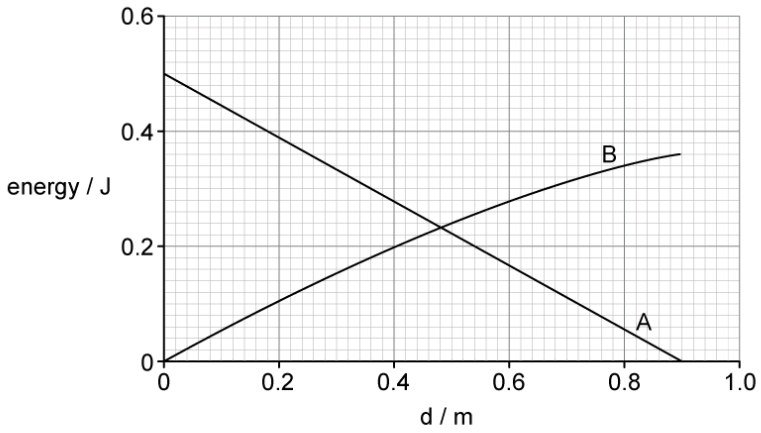


Fig. 1.2

(i) On Fig. 1.2, identify the line representing gravitational potential energy with **GPE** and the line representing kinetic energy with **KE**.

[2]

(ii) Line **A** is linear and line **B** is not linear. Use your answer to part **(i)** to explain why.

[2]

Part (c) [2 marks]

Use the graph in Fig. 1.2. to estimate the energy transferred to thermal energy.

Part (d) **[3 marks]**

Determine the average resistive force acting on the toy car.

Question 9 Medium [13 marks]

Part (a) [4 marks]

A cyclist of mass 80 kg riding a bicycle of mass 12 kg freewheels from rest 650 m down a hill.

The foot of the hill is 20 m lower than the cyclist's starting point and the cyclist reaches a top speed of 12 m s^{-1} before they start to brake at the foot of the hill.

For this situation, calculate

(i) the change in gravitational potential energy

[2]

(ii) the kinetic energy gained by the cyclist.

[2]

Part (b) [2 marks]

For the cyclist and bicycle

(i) State the main resistive forces to the motion.

[1]

(ii) Calculate the work done against these forces during the descent.

[1]

Part (c) [2 marks]

Calculate the average resistive force acting on the cyclist during the descent.

Part (d) **[5 marks]**

The cyclist buys a set of tyres which claim to 'increase efficiency by 30% compared to your old tyres'.

The cyclist tests out the new tyres by riding down the same hill. With the new tyres, the cyclist achieves a top speed of 14 m s^{-1} at the bottom of the hill.

Deduce whether the claim is correct or incorrect.

Question 10 Hard [11 marks]

Part (a) [5 marks]

It has been predicted that in the future, large offshore wind turbines may have a power output ten times that of the largest ones currently in use. These turbines could have a blade length of 50 m or more. A turbine such as this is shown in Fig. 1.1 below.

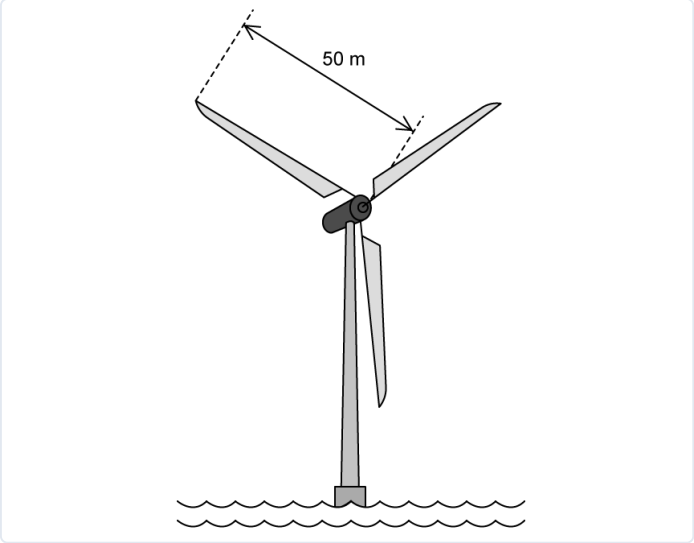


Fig. 1.1

Calculate the power this turbine can produce, at a wind speed of 15 m s^{-1} assuming that it is 100% efficient.

Give your answer in MW.

The density of air is 1.2 kg m^{-3}.

Part (b) [4 marks]

A factory uses 14 100 kWh of electricity per year.

Determine the number of factories that the wind turbine from part (a) can power for 1 year.

Assume that the wind is blowing onto the wind turbine at 15 m s^{-1} on average, for 24 hours every day, 365 days a year.

Part (c) [2 marks]

A German physicist Albert Betz concluded in 1919 that no wind turbine can convert more than 59.3 % of the kinetic energy of the wind into mechanical energy turning the rotor, which then becomes electrical energy. This is known as the Betz limit.

The coefficient of power, C_p of a wind turbine is a measure of how efficiently the wind turbine converts the energy in the wind into electricity. A good wind turbine has a C_p that reaches 70 % of the Betz limit.

Assuming the wind turbine in part (a) is considered a good turbine, calculate its revised power output.

Question 11 Hard [10 marks]

Part (a) [3 marks]

A cyclist rides along a road up an incline at a steady speed of 5.0 m s^{-1}. The mass of the rider and bicycle is 70 kg and the bicycle travels 27 m along the road for every 2.0 m gained in height. Neglect energy loss due to frictional forces.

Calculate the power developed by the cyclist in riding up the slope.

Part (b) [4 marks]

The cyclist now travels along a steeper incline at an angle of 6.5° to the horizontal.

Calculate the power output required from the cyclist on this new plane and discuss

how this value differs from that in part (a).

Assume the cyclist maintains the same speed along the road.

Part (c) [3 marks]

The cyclist stops pedalling and the bicycle freewheels up the incline for a short time.

Considering the energy transfers, calculate the distance travelled along the slope from when the cyclist stops pedalling to where the bicycle comes to rest.